

Customer Churn Prediction & Lifetime Value (LTV) Engine

Key Insights

1. Customer churn is highest among new customers

Customers with shorter tenure were much more likely to leave the service. This shows that the first few months are critical, and improving the onboarding experience could help reduce early churn.

2. Higher monthly charges are linked to higher churn

Customers paying higher monthly fees showed a greater tendency to churn. This suggests that pricing, perceived value, or service satisfaction may be influencing their decision to leave.

3. Customers with longer tenure are more loyal

The analysis revealed that customers who have stayed with the company for a longer period are much less likely to churn. Long-term customers contribute significantly to overall business stability.

4. Contract type has a major impact on retention

Customers with month-to-month contracts had the highest churn rate, while those with one-year and two-year contracts were far more likely to remain with the company. Longer contracts appear to encourage customer loyalty.

5. Electronic check users are more likely to churn

Among the different payment methods, customers using electronic checks showed the highest churn rate. This could indicate lower customer engagement or a preference pattern that deserves further investigation.

6. Customers using multiple additional services are less likely to leave

Customers who subscribed to services such as Online Security, Tech Support, Device Protection, and Online Backup generally had lower churn rates. These services seem to increase customer satisfaction and long-term retention.

7. Internet service type influences customer behaviour

Customers using Fiber Optic internet experienced higher churn compared to those using DSL. Although Fiber Optic offers higher speeds, customer expectations may also be higher, leading to increased dissatisfaction when issues occur.

8. Most churned customers are non-senior citizens

The dashboard shows that the majority of customers who churned are non-senior citizens. Since non-senior customers make up a larger portion of the customer base, they also account for a higher number of churn cases. This highlights the importance of implementing retention strategies across all customer segments rather than focusing on a single age group.

9. High-value customers deserve special attention

Customers with higher predicted Lifetime Value (LTV) represent the most valuable segment of the business. Retaining these customers should be a priority through personalized offers, loyalty programs, and proactive customer support.

10. Machine learning enables proactive retention strategies

The churn prediction model helps identify customers who are at risk of leaving before they actually churn. This allows the company to take preventive actions such as targeted discounts, improved customer support, or personalized engagement campaigns.

Business Recommendations

- Focus on improving the onboarding experience for new customers.
- Offer attractive incentives for customers to move from month-to-month contracts to long-term plans.
- Review pricing strategies for customers with high monthly charges.
- Encourage customers to adopt value-added services such as Online Security and Tech Support.
- Design personalized retention campaigns for customers identified as high churn risk.
- Provide exclusive benefits and loyalty rewards for customers with high Lifetime Value.
- Continuously monitor churn trends using the Power BI dashboard to support data-driven decision making.

Conclusion

This project demonstrates how data analytics, machine learning, and business intelligence can work together to solve a real business problem. By predicting customer churn, estimating customer lifetime value, and presenting insights through an interactive Power BI dashboard, the

project helps businesses identify at-risk customers, understand the key factors behind churn, and make informed decisions to improve customer retention and long-term profitability.